15/10/2026

Hooke's Law - Determination of the Spring Constant k

(1) Objective :-

Calculating the Spring Constant (k).

(2) Law used :-

$$m.g = k.\Delta e$$

m ⟶ suspended mass (gm)

g ⟶ acceleration of gravity = 980 cm/s²

k ⟶ Spring Constant (dyne/cm)

Δl ⟶ extension = $l_1 - l_0$ (cm)

(3) Experimental data

l_0 = ---- cm

m (gm)	Δl (cm)

(4) Slope & results

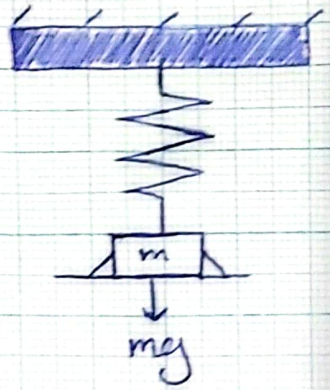
Slope = Δ(Δl) / Δm = ...

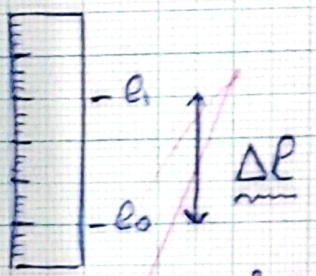
k = g / slope = 980 / slope

[k = ---- dyne/cm]

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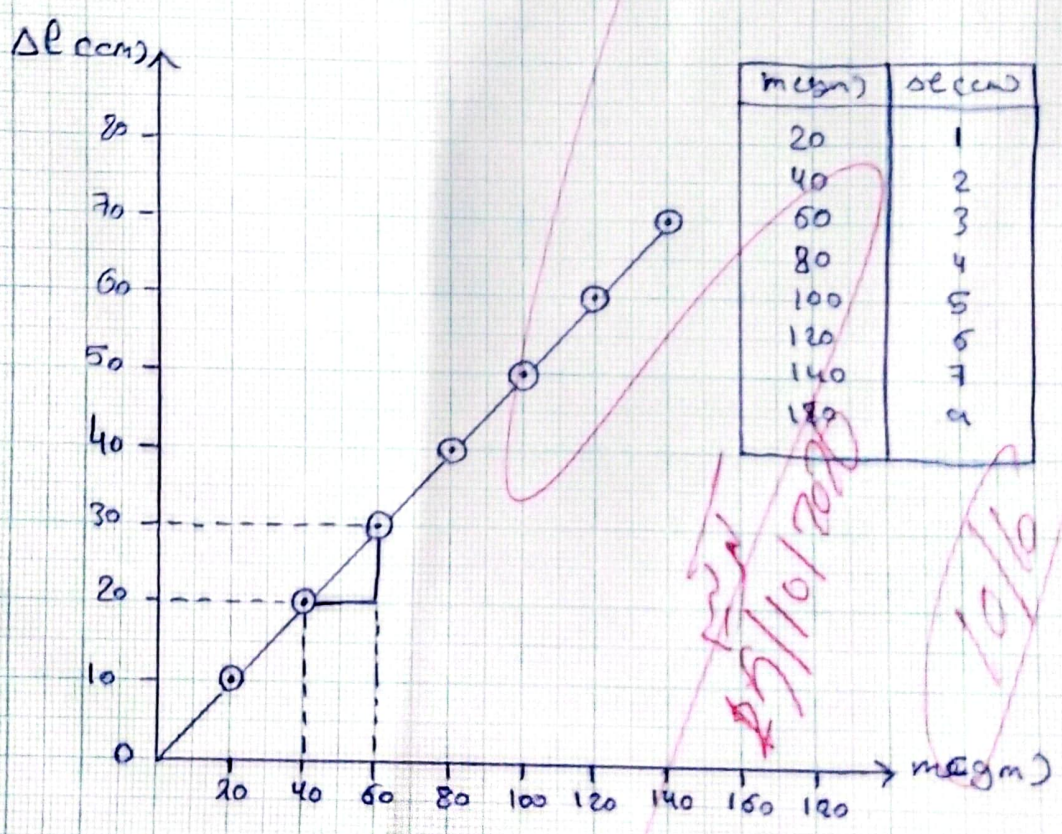
① Diagram





Scale (cm)

② Graph : $\Delta\ell$ vs m



m (gm)	Δl (cm)
20	1
40	2
60	3
80	4
100	5
120	6
140	7
160	8

$$\text{Slope} = \frac{30-20}{60-40} = \frac{10}{20} = 1/2$$

$$k = 980/\text{slope} = 980 \times 2 = 1960 \ \frac{\text{dyne}}{\text{cm}}$$

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